

Issue # 7: Numerator Data - Descriptive Statistics 2021-2022

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This Write-up contains tables and plots for the descriptive analysis of the Numerator data. The original user-trip-item level data is transformed to user-month level data, with variables summarized at the monthly level for each user.

1. Tables

- FAH/FAFH department codes (less granular grouping) and soda category (more granular grouping) codes
- Number of users by state-year
- Demographics by treatment status
- Spending by treatment status
- Soda spending by treatment status
- Food and soda spending percentiles

2. Figures: Distribution of ...

- proportion of food spending out of total spending
- proportion of FAH/FAFH spending out of total food spending
- proportion of FAH/FAFH on SNAP out of SNAP spending
- soda spending/winsorized spending
- Proportion of soda spending out of total food spending
- proportion of soda on SNAP out of total SNAP spending
- proportion soda at home/away from home

Table 1: Food at Home vs. Food Away from Home Categories

Food At Home (Grocery)	Food Away From Home (QSR)
Snack	LSR Sandwiches & Wraps
Canned	LSR Beverages
Produce	LSR Breakfast
Dairy	LSR Desserts
Beverages	LSR Entrees
Packaged Bakery (Bread & Alternatives)	LSR Snack & Sides
Condiments	LSR Mexican
Frozen Foods	LSR Sauces & Condiments
Shelf Stable Meals	LSR Italian
Candy (Snacks)	LSR Salads
Baking & Cooking	
Meat	
Seafood & Fish	
Beans & Grains	
Refrigerated Foods	
Breakfast	
Pasta & Noodles	
Deli & Prepared Foods	
In-Store Bakery (Bread & Alternatives)	
Herbs & Spices	
Alcohol Beverages	
Misc. Food	
Fresh Meal Kits	
Ice	

Table 2: Soda Categories

Soda At Home (Grocery Beverages)	Soda Away From Home (QSR Beverages)
Soft Drinks	LSR Bottled Soda
Tonic & Club Soda	LSR Fountain Drink
Sparkling Juice & Ciders	
Prebiotic & Probiotic Soda	

Table 3: Number of Users in Affected and Unaffected States

State	2021	2022
Household affected in Jan 2026		
FL	20910	21461
IA	2658	2714
ID	1286	1305
IN	6426	6548
LA	3714	3801
NE	1747	1775
OK	3091	3138
UT	1991	2023
WV	2005	2037
Subtotal	43828	44802
Household affected after Jan 2026		
AK	497	504
CO	3731	3810
TX	22382	22857
Subtotal	26610	27171
Treated States	70438	71973
Control States	195023	198786
Total	265461	270759

Table 4: User-Month Panel - Demographics

	Treated States	Mean (SD) Control States	Total	Min	Max
Income	79271.59 (58651.85)	84611.78 (63127.45)	83731.09 (62442.90)	10000.00	275000.00
Household Size	2.85 (1.47)	2.91 (1.49)	2.90 (1.49)	1.00	7.00
Age	50.30 (12.64)	48.80 (12.44)	49.05 (12.49)	19.00	67.50
Married	0.57 (0.49)	0.55 (0.50)	0.55 (0.50)	0.00	1.00
Education (years)	13.80 (1.34)	13.86 (1.37)	13.85 (1.36)	10.00	16.00
White	0.76 (0.43)	0.69 (0.46)	0.70 (0.46)	0.00	1.00
Black	0.08 (0.27)	0.10 (0.30)	0.10 (0.30)	0.00	1.00
Asian	0.03 (0.17)	0.08 (0.27)	0.07 (0.26)	0.00	1.00
Hispanic	0.11 (0.32)	0.11 (0.31)	0.11 (0.31)	0.00	1.00
Ever Child	0.26 (0.44)	0.27 (0.45)	0.27 (0.44)	0.00	1.00
Number of States	9	48	57		
Number of Users	44,863	226,343	271,206		
Average Number of Months per User	22.69	22.78	22.76		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 5: User-Month Panel - Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Food Spending	455.62 (399.95)	459.10 (411.60)	458.52 (409.71)	0.00	19654.96
Total Spending	1190.43 (1026.53)	1190.13 (1026.81)	1190.18 (1026.76)	0.00	69786.66
Prop. of Total Spending on Food	0.40 (0.22)	0.40 (0.22)	0.40 (0.22)	0.00	1.00
Prop. of Food Spending on FAH	0.92 (0.14)	0.91 (0.15)	0.91 (0.15)	0.00	1.00
Prop. of Food Spending on FAFH	0.08 (0.14)	0.09 (0.15)	0.09 (0.15)	0.00	1.00
Prop of Food on EBT	0.39 (0.29)	0.40 (0.29)	0.40 (0.29)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.02)	1.00 (0.03)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.02)	0.00 (0.03)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.16 (0.37)	0.17 (0.38)	0.17 (0.38)	0.00	1.00
Number of States	9	48	57		
Number of Users	44,863	226,343	271,206		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 6: User-Month Panel - Soda Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Soda Spending	17.38 (29.40)	16.03 (29.24)	16.25 (29.27)	0.00	5800.09
Prop. of Food Spending on Soda	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.90 (0.24)	0.90 (0.25)	0.90 (0.25)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.10 (0.24)	0.10 (0.25)	0.10 (0.25)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.11)	0.05 (0.10)	0.05 (0.11)	0.00	1.00
Prop of Soda Spending on EBT	0.36 (0.37)	0.37 (0.38)	0.37 (0.38)	0.00	1.00
Number of Users	44,863	226,343	271,206		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 7: User-Month Panel - Demographics (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Income	50441.27 (49410.31)	89388.23 (62661.87)	83731.09 (62442.90)	10000.00	275000.00
Household Size	3.13 (1.68)	2.86 (1.45)	2.90 (1.49)	1.00	7.00
Age	48.86 (12.10)	49.08 (12.55)	49.05 (12.49)	19.00	67.50
Married	0.33 (0.47)	0.59 (0.49)	0.55 (0.50)	0.00	1.00
Education (years)	13.38 (1.42)	13.93 (1.34)	13.85 (1.36)	10.00	16.00
White	0.55 (0.50)	0.73 (0.45)	0.70 (0.46)	0.00	1.00
Black	0.19 (0.39)	0.08 (0.28)	0.10 (0.30)	0.00	1.00
Asian	0.06 (0.23)	0.07 (0.26)	0.07 (0.26)	0.00	1.00
Hispanic	0.18 (0.38)	0.10 (0.30)	0.11 (0.31)	0.00	1.00
Ever Child	0.36 (0.48)	0.26 (0.44)	0.27 (0.44)	0.00	1.00
Number of States	53	57	57		
Number of Users	40,067	231,139	271,206		
Average Number of Months per User	22.38	22.83	22.76		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 8: User-Month Panel - Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Food Spending	588.70 (477.36)	436.40 (392.80)	458.52 (409.71)	0.00	19654.96
Total Spending	1370.52 (1104.12)	1159.53 (1009.84)	1190.18 (1026.76)	0.00	69786.66
Prop. of Total Spending on Food	0.46 (0.20)	0.39 (0.22)	0.40 (0.22)	0.00	1.00
Prop. of Food Spending on FAH	0.91 (0.13)	0.91 (0.15)	0.91 (0.15)	0.00	1.00
Prop. of Food Spending on FAFH	0.09 (0.13)	0.09 (0.15)	0.09 (0.15)	0.00	1.00
Prop of Food on EBT	0.46 (0.28)	0.28 (0.26)	0.40 (0.29)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.02)	1.00 (0.03)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.02)	0.00 (0.03)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.78 (0.41)	0.07 (0.25)	0.17 (0.38)	0.00	1.00
Number of States	53	57	57		
Number of Users	40,067	231,139	271,206		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization. Food spending and total spending are higher for SNAP than Non-SNAP household, potentially reflecting the consequence of Numerator's monetary incentives. Household on SNAP, who tend to earn less income, tend to upload more receipt with larger values (incentives vary based on receipt value).

Table 9: User-Month Panel - Soda Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Soda Spending	25.32 (36.65)	14.71 (27.53)	16.25 (29.27)	0.00	5800.09
Prop. of Food Spending on Soda	0.05 (0.06)	0.04 (0.06)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.91 (0.21)	0.89 (0.25)	0.90 (0.25)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.09 (0.21)	0.11 (0.25)	0.10 (0.25)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.09)	0.05 (0.13)	0.05 (0.11)	0.00	1.00
Prop of Soda Spending on EBT	0.43 (0.38)	0.25 (0.34)	0.37 (0.38)	0.00	1.00
Number of Users	40,067	231,139	271,206		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 10: User-Month Panel - Food Spending Percentiles

percentile	food_spending_2021	food_spending_2022
50	343.82	420.42
60	433.47	515.97
70	536.86	627.13
80	669.44	770.38
90	879.73	1000.16
95	1085.77	1226.30
96	1153.45	1300.89
97	1243.25	1399.06
98	1375.94	1545.00
99	1630.48	1824.84
99.5	1951.74	2182.38
100	19265.99	19654.96

Table 11: User-Month Panel - Soda Spending Percentiles

percentile	soda_spending_2021	soda_spending_2022
50	4.79	7.18
60	9.07	12.48
70	14.95	19.36
80	23.96	29.92
90	40.96	49.57
95	59.85	71.36
96	66.31	78.70
97	74.90	88.58
98	87.64	103.07
99	110.80	129.77
99.5	136.46	159.57
100	3389.28	5800.09

Figure 1: Proportion of Food Spending

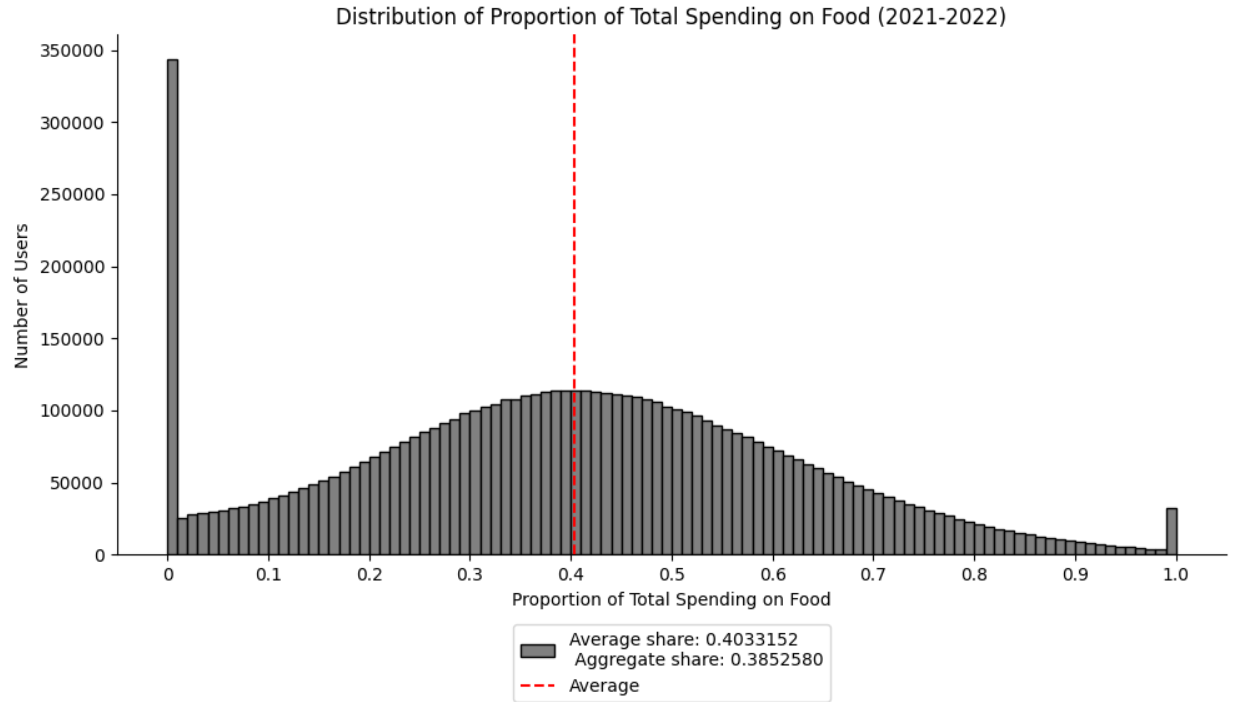


Figure 2: Proportion of FAH/FAFH Spending out of total food spending

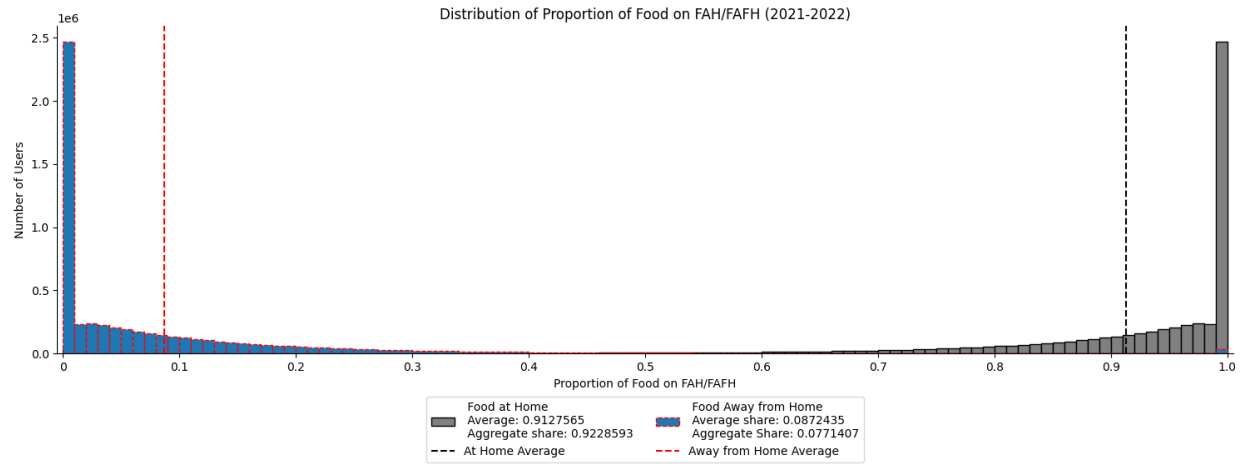
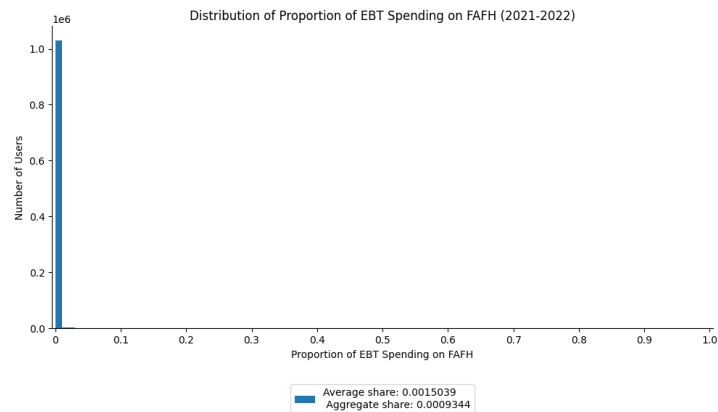


Figure 3: Proportion of FAH/FAFH SNAP Spending out of total SNAP spending

(a) Food Away



(b) Food at Home

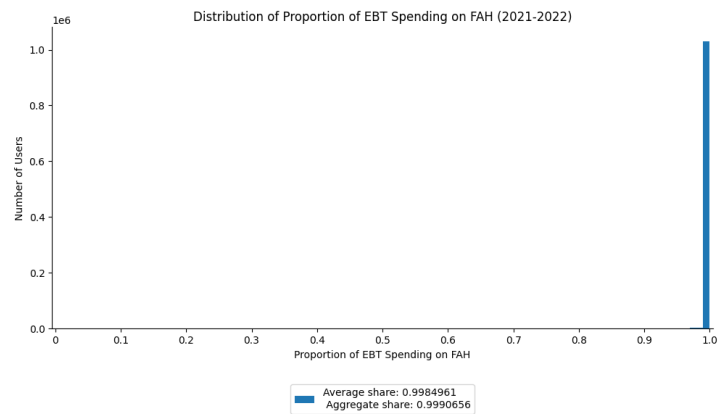
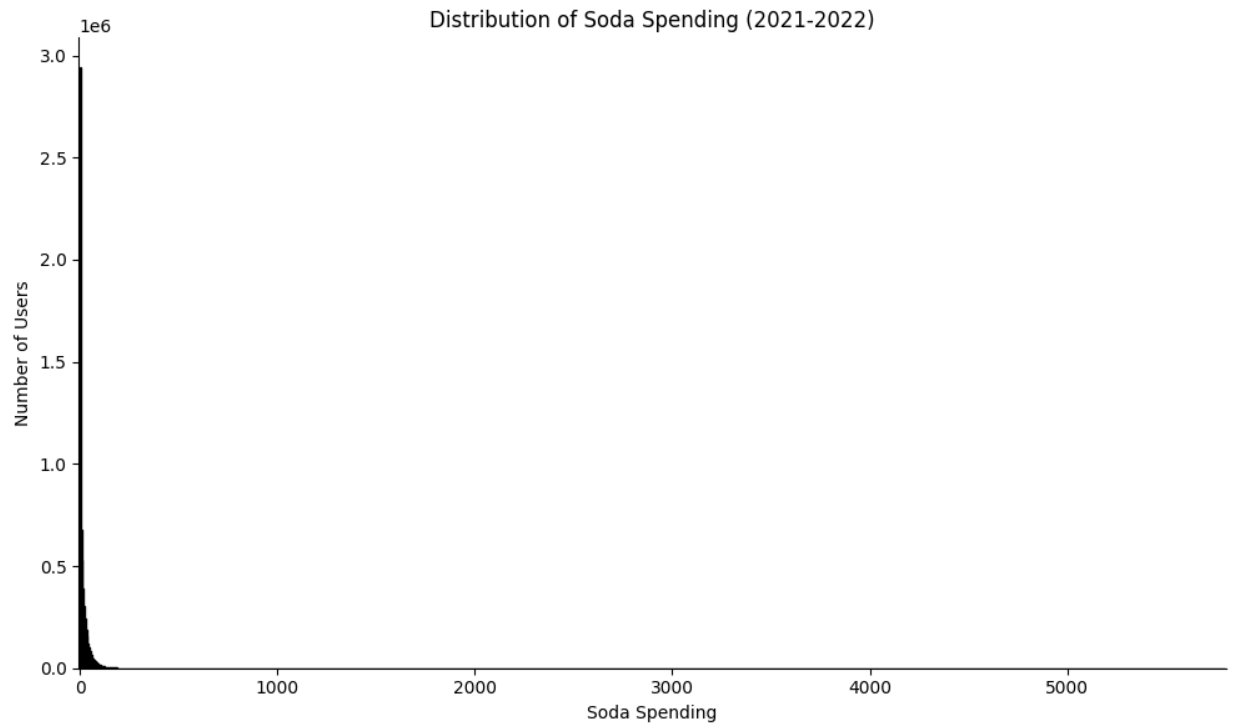


Figure 4: Soda Spending

(a) unwinsorized



(b) winsorized at 99.5%

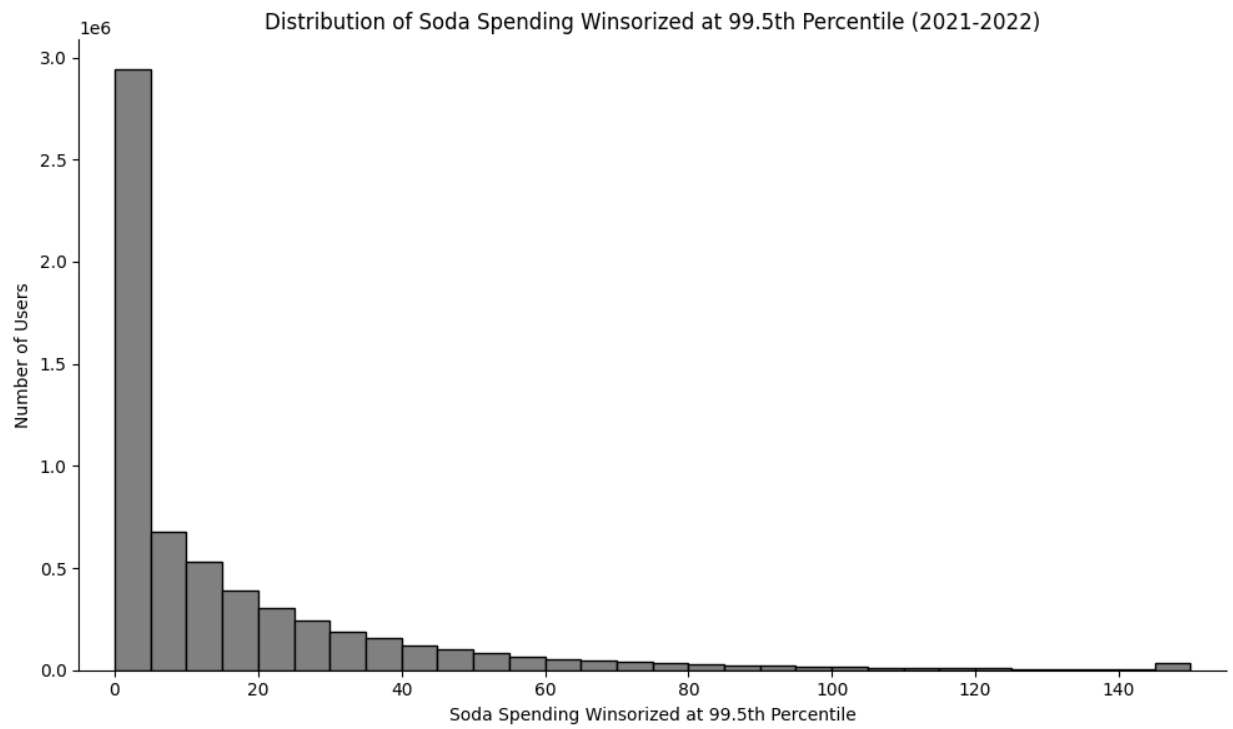


Figure 5: Proportion of Soda Spending out of total food spending

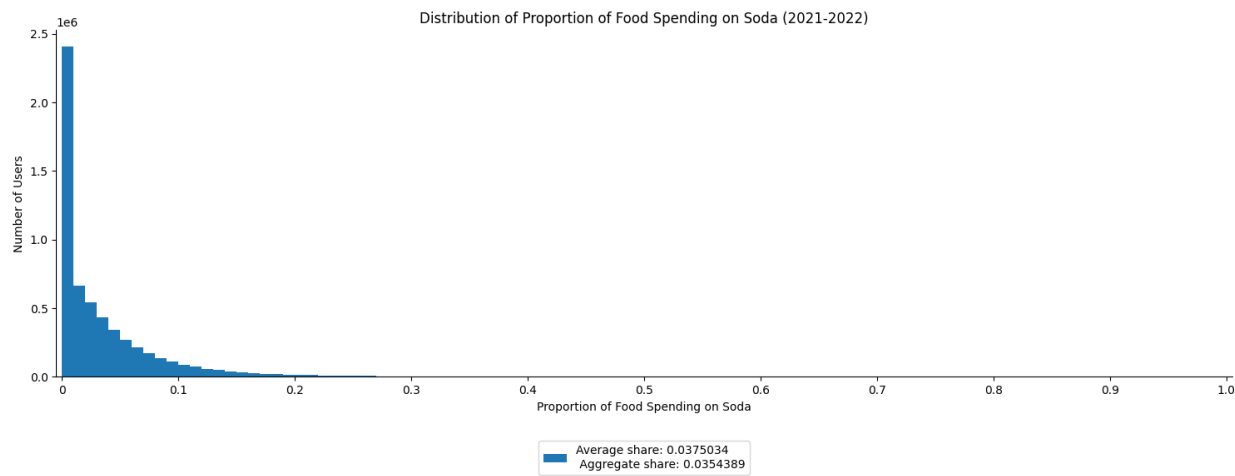


Figure 6: Proportion of Soda SNAP Spending out of total SNAP spending

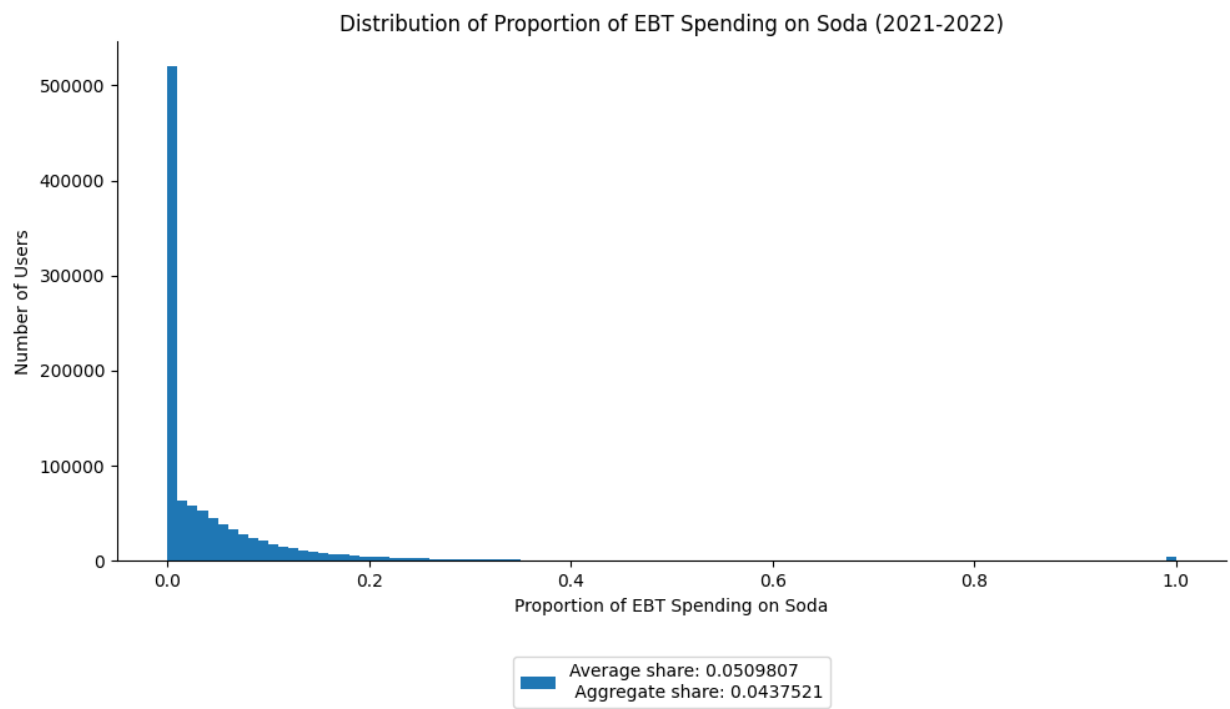


Figure 7: Proportion of Soda FAH/FAFH Spending out of total soda spending

